time taken, $t_2 - t_1 = 0.8000 - 0.5035 = 0.2965 = 0.30$ s

**13** **C** From the graph, $x_0 = 0.50$ m and $v_0 = 2.0$ m s$^{-1}$

$$v_0 = \omega x_0 \quad \Rightarrow \quad \omega = \frac{v_0}{x_0} = \frac{2.0}{0.50} = 4.0 \text{ rad s}^{-1}$$

total mechanical energy, $E_T = \frac{1}{2} m\omega^2 {x_0}^2$

Since $E_T$ and $m$ are constant, $\omega^2 {x_0}^2$ must remain constant.

$$\omega' x_0' = \omega x_0 \quad \Rightarrow \omega' = \frac{4.0 \times 0.50}{0.25} = 8.0 \text{ rad s}^{-1}$$

$$|a_0| = (\omega')^2 x_0' = 8.0^2 \times 0.25 = 16 \text{ m s}^{-2}$$

**14** **D** $\lambda = 16$ m from diagram (C to C or R to R)